Output Analysis

Systems Modeling and Simulation

4DV650

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Output analysis

- Simulation's purpose is either to **predict the performance** of a system or to **compare the performance** of two or more alternate system designs

- The need for **statistical output analysis** is based on the observation that the output data from a simulation exhibits **random variability**
  - Due to use of random numbers to produce input variables
    - Two different streams or sequences of random variables will produce two sets of outputs which will differ

- Output analysis is the **examination** of the data generated by a simulation

Statistical Output Analysis

- **Objective**: *Estimate system performance via simulation*
- If the system performance is measured by θ, the result of a set of simulation experiments will be an **estimator** $\hat{\theta}$ of θ
- The **precision** of the estimator $\hat{\theta}$ can be measured by:
  - The **standard error** of $\hat{\theta}$
  - The width of a **Confidence Interval** (CI) for θ.
- Purpose of statistical analysis:
  - To estimate the **standard error** or **CI**
  - To figure out the **number of observations** required to achieve desired error/CI

Types of Simulation w.r.t. Output Analysis

- Depends on both the **objectives** of the simulation study and the **nature** of the system
- **Terminating simulation (transient):**
  - Runs for some duration of time T_E
    - where E is a specified event that stops the simulation
  - Starts at time *0* under initial conditions and ends at the stopping time T_E
    - T_E is determined by the problem
- **Non-terminating simulation (steady-state)**
  - Runs **continuously**, or at least over a very long period of time
  - Starts at time *0* under initial conditions and ends at the stopping time T_E
    - T_E is set by the analyst as one parameter in the design of the simulation experiment
  - Used to study **steady-state** (long-run) properties of the system
    - Properties that are **not influenced** by the initial conditions of the model at time 0

Output Analysis for Terminating Simulations

- A Simulation running over the interval $[0, T_E]$ and resulting in observations $Y_1, \ldots, Y_n$
- **Independent replications**

  The simulation is repeated a total of R times, each run using a different random number stream and independently chosen initial conditions

Within Replication Data	Across Replication Data
$Y_{11}\ Y_{12}\ \ldots\ldots\ Y_{1n1}$	$\bar{Y}_{1.}, S_1^2, H_1$
$Y_{21}\ Y_{22}\ \ldots\ldots\ Y_{2n2}$	$\bar{Y}_{2.}, S_2^2, H_2$
…	
$Y_{R1}\ Y_{R2}\ \ldots\ldots\ Y_{RnR}$	$\bar{Y}_{R.}, S_R^2, H_R$

?

$$\bar{Y}_{..}, S^2, H$$

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

$$H_i = t_{\alpha/2, n_i - 1}\frac{S_i}{\sqrt{n_i}}$$

H is confidence interval half-width

Across-replication (discrete-time) data

Across-replication data are independent (different random numbers) and identically distributed (same model)

- Sample average:

$$\bar{Y}_{..} = \frac{1}{R}\sum_{i=1}^{R} Y_{i.}$$

- Sample variance:

$$S^2 = \frac{1}{R-1}\sum_{i=1}^{R} (Y_{i.} - \bar{Y}_{..})^2$$

- Confidence-interval half-width:

$$H = t_{\alpha/2,R-1}\frac{S}{\sqrt{R}}$$

Within Replication Data	Across Replication Data
Y_{11} Y_{12} ..... Y_{1n1}	$\bar{Y}_{1.}, S_1^2, H_1$
Y_{21} Y_{22} ..... Y_{2n2}	$\bar{Y}_{2.}, S_2^2, H_2$
...	
Y_{R1} Y_{R2} ..... Y_{RnR}	$\bar{Y}_{R.}, S_R^2, H_R$

*$t_{\alpha/2,R-1}$ *from* t distribution from Table A.5 in the Appendix

Confidence Intervals with Specified Precision

- The half-length H is given by: $H = t_{\alpha/2,R-1}\frac{S}{\sqrt{R}}$ → Standard error

1. Assume that an initial sample of size R_0 replications has been observed
2. Obtain an initial estimate of the sample variance S_0^2
3. Choose sample size R such that $R \geq R_0$ *and*

$$H = t_{\alpha/2,R-1}\frac{S_0}{\sqrt{R}} \leq \epsilon$$ → Error criterion

4. Collect R - R_0 additional observations
5. Form the *100(1-*α*)%* Confidence interval for θ:

$$\bar{Y}_{..} \pm t_{\alpha/2,R-1}\frac{S}{\sqrt{R}}$$

Output Analysis for Steady-State Simulation

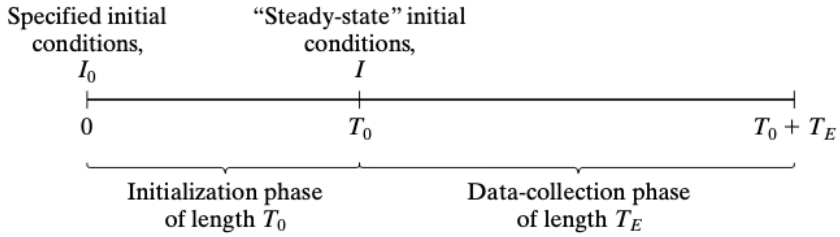
- The steady-state measure of performance is defined by $\theta = \lim_{n \to \infty} \frac{1}{n} \sum_{i=1}^{n} Y_i$
- The sample size (*n or* T_E) is a design choice, considering:
  - Any **bias** in the point estimator that is due to artificial or arbitrary initial conditions
    - bias can be severe if run length is too short
  - Desired **precision** of the point estimator
- Methods to reduce the point-estimator bias
  - **Intelligent initialization**
  - Divide simulation into an **initialization phase** and **data-collection phase**

Intelligent initialization

1. If the system exists, collect data on it and use these data to specify typical initial conditions
2. If the systems doesn't exist
   - Solve the mathematical model and make use of the results to initialize the simulation

Initialization Bias in Steady-State Simulations

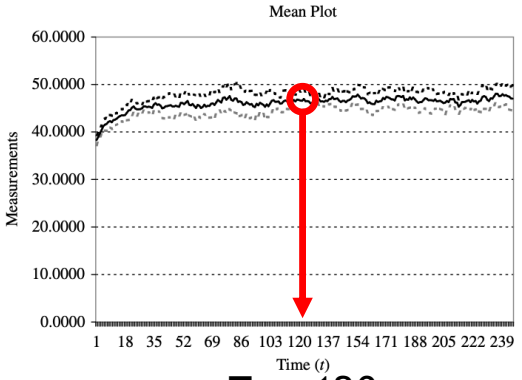
- Divide each simulation into two phases:
  - An **initialization phase**, from time *0* to time T_0.
  - A **data-collection phase**, from T_0 to the stopping time T_0+T_E.



  - The choice of T_0 is important, as the system state at time T_0, denoted by I, should be more nearly representative of steady-state behavior than are the original initial conditions I_0 at time 0.
  - **Ensemble averages** technique

Within Replication Data				Across Replication Data
Y_{11}	Y_{12}	.....	Y_{1n1}	
Y_{21}	Y_{22}	.....	Y_{2n2}	
...				
Y_{R1}	Y_{R2}	.....	Y_{RnR}	

$$\bar{Y}_{\cdot j} = \frac{1}{R}\sum_{r=1}^{R} Y_{rj}$$



$T_0 = 120$

Replication Method for Steady-State Simulation

- If initialization bias has been **reduced**, then the method of **independent replications** can be used to calculate the point-estimator and to **construct a confidence interval**

Approach:

1. make R replications
2. initializing and deleting
3. Calculate Replication Averages:

$$\bar{Y}_{r.}(n,d) = \frac{1}{n-d}\sum_{j=d+1}^{n} Y_{rj}$$

4. Calculate the overall point estimator

$$\bar{Y}_{..}(n,d) = \frac{1}{R}\sum_{r=1}^{R} \bar{Y}_{r.}(n,d)$$

Replication	Observations						Replication Averages
	1	...	d	d+1	...	n	
1	~~$Y_{1,1}$~~	...	~~$Y_{1,d}$~~	$Y_{1,d+1}$	...	$Y_{1,n}$	
2	~~$Y_{2,1}$~~	...	~~$Y_{2,d}$~~	$Y_{2,d+1}$	...	$Y_{2,n}$	
.	.		.	.		.	
.	.		.	.		.	
.	.		.	.		.	
R	~~$Y_{R,1}$~~	...	~~$Y_{R,d}$~~	$Y_{R,d+1}$	...	$Y_{R,n}$	

5. Estimate **standard error** by means of sample variance

$$S^2 = \frac{1}{R-1}\sum_{r=1}^{R}(\bar{Y}_{r.} - \bar{Y}_{..})^2 = \frac{1}{R-1}\left(\sum_{r=1}^{R}\bar{Y}_{r.}^{\,2} - R\bar{Y}_{..}^{\,2}\right) \quad \text{and} \quad s.e.(\bar{Y}_{..}) = \frac{S}{\sqrt{R}}$$

As a rough rule:

- TE > $10T_0$
- R <= 25 replications

Materials (MyMoodle)

- Reading
  - S3 [Lecture] S3 [Lecture] Output Analysis - Reading.pdf

- Exercises (with solutions):
  - S3 [Lecture] Output Analysis - Exercises.pdf
  - S3 [Lecture] Output Analysis – Reading - Solutions.pdf